Page No.
Date

Unit : 3

Cloud computing platform and cloud native technologies

- Cloud computing utilizes:

Cloud computing utilizes the framework i.e build for hosting extensive distributed network applications on the internet.

Cloud Architecture enables the integration of software operating on virtualize hardware across several location to deliver a user centric service on demand.

Compability:

Cloud based applications typically the characteristics of being from a set of components which are known as compatibility. A composible system utilizes components to provide service that can be customize for a specific purpose using standardized elements. A composible component must the following characters.

1) Modular

A modular unit a self contain and independant component i.e. composable, reusable and replacable.

2) Stateless

A Transaction is conducted independantly of other transaction or request.

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Page No.
Date

Infrastructure

The majority of major infrastructure and service providers depend on virtual machine technologies to supply services capable of running applications.

The virtual machine component of a hypervisor is the virtual machine monitor. This software is responsible for enabling several or. To operate independently own allocated memory and handles Input / output for the virtual machine.

Platforms:

A cloud platform is a software that performs the creation of more advance services various platform as a services vendors offer services design to give developers with device capability.

platform encompasses almost the whole cloude services with the exception of the presentation layer responsible for user interface

platform frequently includes a wide range of tools and utilizes that assist in the process of designing and deploying applications various vendors offer to team, communication testing program performance measurement attribute analysis database and webservice integration and storage management.

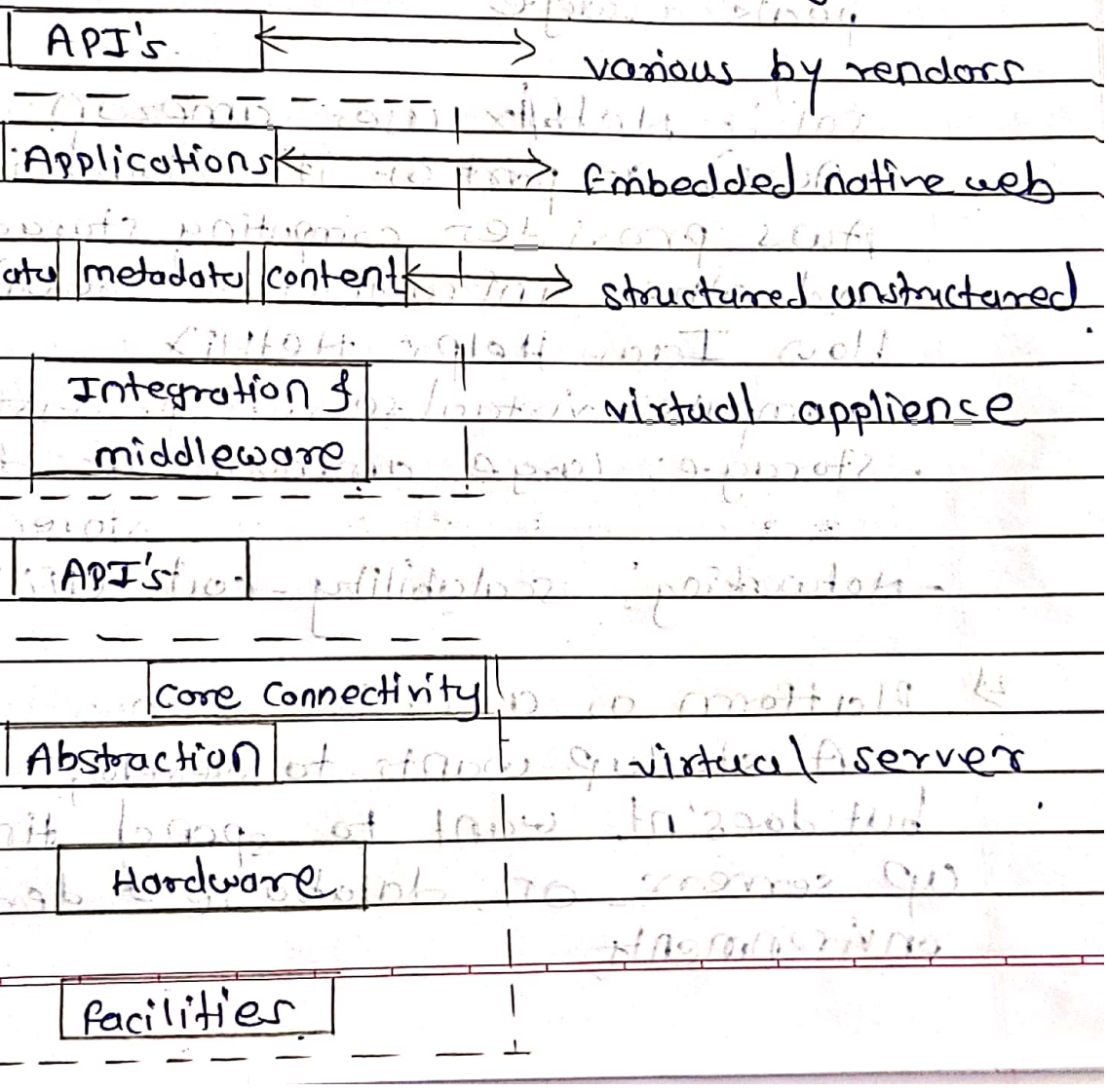
Typically platform is initiate the process by creating a developer community to provide assistance and support for the task platform within the system.

Virtual appliances:

it is are applications such as web servers or database servers that can be excited on a virtual machine image

A virtual appliances are software program that are loaded of virtual servers. They are design to execute on specific machine instances or image type

A virtual appliances is an instance of a platform Virtual appliances are position in the middle of Cloud computing.



Page No.
Date

Platform as
- Case study on

1) Infrastructure as a service
Netflix is popular online streaming platform that provides movies and TV shows to millions of users worldwide. Handling such a large number of user requires huge computing storage and networking resources.
problem:
Netflix needed:
- large amount of storage for video
- powerful servers to handle midleince of users.
- Ability to increase resources during peak usage.

sol^n:- Netflix uses amazon web services (AWS) as part of its cloud infrastructure. AWS provides computing storage networking and other infrastructure resources on demand.
How Iaas Helps Netflix
- compute: virtual servers process application
- storage: large ammount of data can be stored
- Networking: scalability - cost efficiency

2) Platform as a service
A startup wants to develop a web appl but doesn't want to spend time setting up servers or, database & development environments,

problem:
-Developers traditionally have to set up servers, Install os, configure development environments manage scaling, maintain infrastructure.

sol^n. The company uses google App engine a PaaS offering from google cloud.

3) Software as a service

A college or company needs application email, document creation spreadsheets pentation, and collaboration.

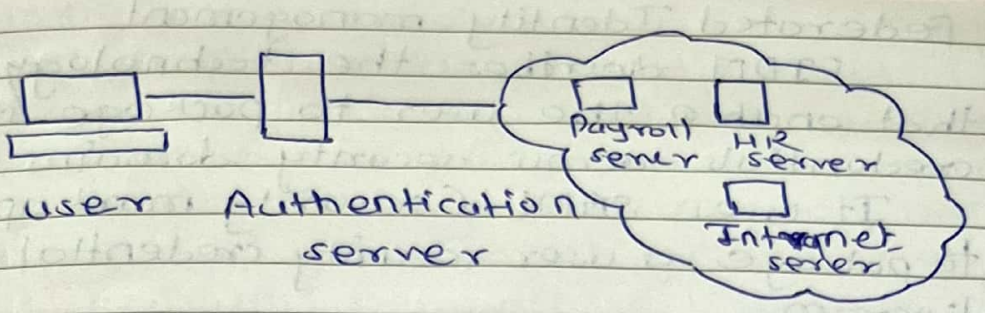
Problem:
Traditionally the organization would be need to purchase software license, Install appl^n on computer, maintain source, perform software updates, manage storage and backups.

sol^n.
The organization uses google workplace which provides application such as gmail google Docs, sheets and slides through the internet.

- Identity as service
  Employers in company require to log in to system to perform various task this systems may be based on local servers or cloud based servers.
  Following other problems that an employee might it face
  i) Remembering username & password for accessing multiple servers.
  ii) if an employee leaves the company it is required that all account to d be disable
  To solve this problem, a new technique called identity as service is been used.

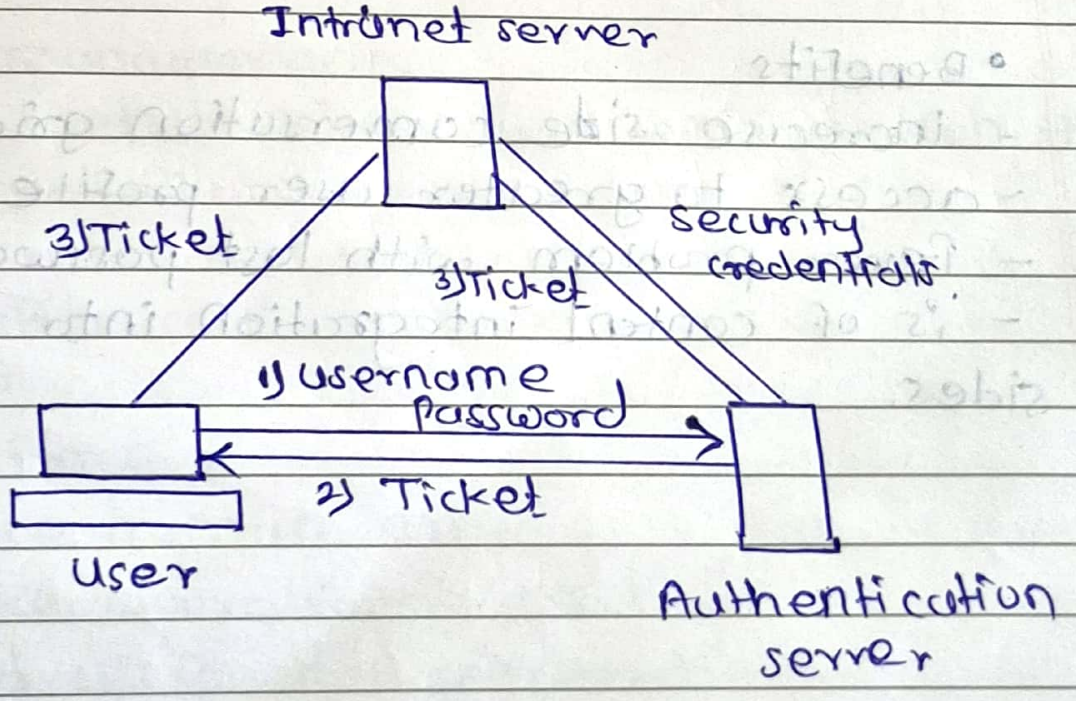
- Identity refers to set of attributes associated with something to make it recognizable. all objects may have same attributes but there identier cannot be same.
  A unique identity is assign through unque identification atribute.

i) single sign on (SSO)
  To solve the problem of using different username & password combination for different servers companies know employee single sign on software which allows the user to log in only one time and manage the access to the other systems.
  - SSO as ~~server~~ single authentication server managing multiple access to other system as shows in diagram.



- SSO working
- Their are separate implementation of SSO following steps explain the working of SSO.

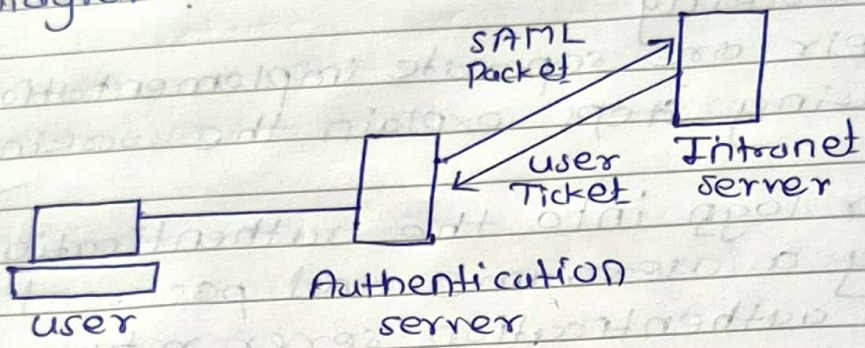
i) user logs into the authentication server using a username and password
ii) The authentication server returns the user ticket.
iii) users sends the ticket to internate server
iv) intranet server to sends the ticket to the authentication server
v) authentication server sends the user security credentials for that servers back to the intranet servers.



2) Federated Identity management

FIDM describes the technology and protocol that enable the user to package security credentials across security domains.

It uses SAML (security mark up language) to package a user security credential as shown in diagram.



3) Open ID

- it offers users to log in into multiple websites with single account.
- Google, Myspace, yahoo, wordpress, dot.com are some of the company that support open ID.

- Benefits
- increase side conversation grid
- access to greater user profile content
- Fewer problem with lost password.
- is of content integration into social networking sides.

- Compliance as service
  - Compliance as service is cloud based delivery model when third party providers or managed service providers automate, manage and monitor an organizations association to industry and government guideliner regulations
  - it performs manuals wi periodic audits into continues automated workflow.

- key features of compliance as service

1) continues monitoring
real time tracking of cloud resource configuration on security controls.

2) Automated evidence gathering
instante of gathering of reports and document required for audit.

3) Policy mapping
Translating complex law into technigual cloud configuration.

4) Risk management
Early detection of miss configuration that violet rules.

- Benefits and droback

- safe time
- Reduce human error.
- lowers operational cost
- speed up audits preparation.

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- Drawbacks
  - creates relience on third party vendor
  - offers limited customization for unique workflow
  - The ultimate legal responsibility for bridge still rest on the business.

- Containers vs virtual machine
  containers and virtual machine are two technologies that enable applications to run in isolated environment

What is container?
- Containers are lightweight portable and self-content executable image that content software applications and they are different dependancy
- They are used to deploy and run applications in consistant way across different environment such as deploy development, staging and production.
- Containers are tipycally deployed from image by using and orchastration platform like google
-

What are virtual machine?
- VM or guest represent instance of an operating system po-located on physical machine through the use of hypervisor.
- each VM has its own operating system memory and other resources, which are isolated from the other VM's on the same physical computer
- This allows multiple operating system to run on the same physical component without interfering with each other

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Page No.
Date

- Difference bet$. container and VM

i) - VM access the hardware of physical machine through a hypervisor. the hypervisor creates an abstraction layer allowing the VM to access CPU, memory, and storage.
- container on the other hand represent a package that includes an executable with dependancies into needs to run this means that each container shares the physical machin hardware and operating system (kernel) with other containers.

ii) - as a result VM's are typically more resource intensive then containers.

Feature	container	virtual machine
1) virtualization level	operating system level	Hardware level
2) operating system	shares host os kernal	guest os
3) size	megabytes	gigabytes
4) Boot time	Boot time for container second	minutes
5) Resourse usage	lower	Higher
6) Isolation	Process level	Hardware level
7) Portability	Highly portable	less portable
8) management	Typically manage by container orchastration tool	manage by hypervisor

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To understand the difference betⁿ containe and vm it is helpful to visulize there are architecture

Containers
i) shares the host kernal
containers leverage the underline operating system kernal of the host machine

ii) lightweight
Smaller in size and requires less resources

iii) faster startup
start and stop quickly you to shared os kernal

Virtual machine
i) Runs a full guest os
each vm has its own dedicated operating system.

ii) Resource intensive
require more resources that is cpu, memory and storage

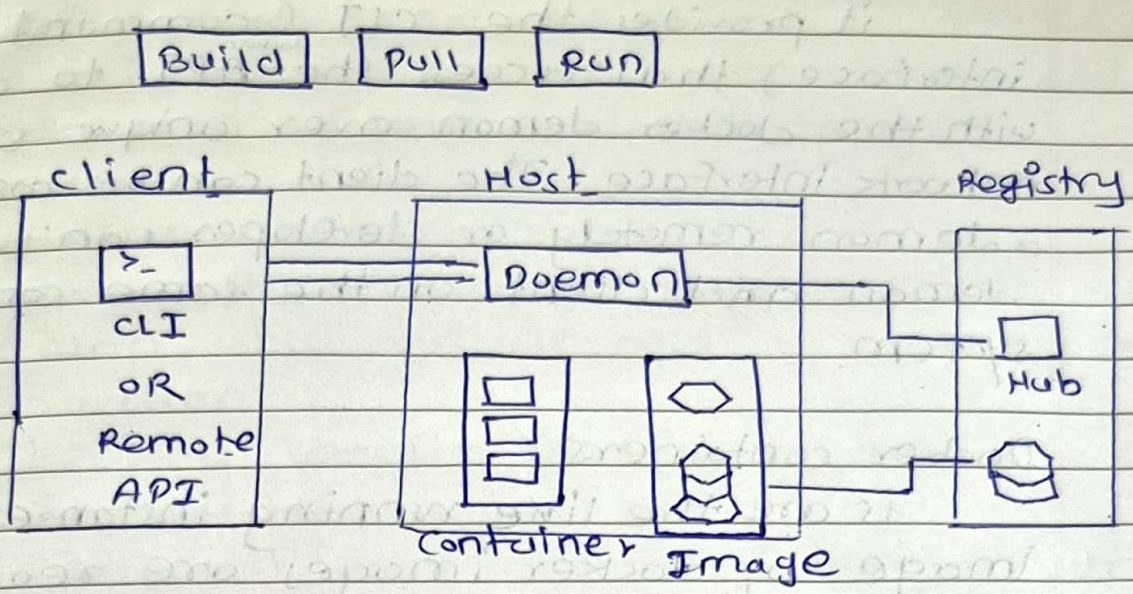
iii) Slower startup
Take longer to boot due to the need to load the guest os.

- Doccur

What is doccur?
A doccur is a open source platform that enables developers to package application and there dependancies into portable containe

that can run consistantly across various environment.

Docurr Architecture.



Docker usage client server architecture following is a break down of the core component associated with docker.

Docker Host

A Docker Host is a physical or virtual machine Running linux or another docker engine compatable os.

Docker Engine

Docker engine is a client server application consisting of the docker doemon. a docker API that interacts with the doemon and command line interface that docs to the doemon.

Docker Doemon

is a service that creates and manages docker images. by using the commands from the client

essentially the docker daemon search as a control centre for the docker implementation.

Docker client
it provider the CLI (command line interface) that access the docker API to communicate with the docker demon over unique socket or network interface. The client can be connected to a demon remotely or developer can run the demon and client on the some computer system.

Docker containers
is are the like running instances of docker image by docker images are read only files

Docker image
A Docker image contain executable appln source code and all the tools, libraries and dependencies the appl code needs to run as a container.

Docker build
is a command that has tools and features for building docker image.

Docker files
each docker container starts with the simple text file containing instructions for how to build docker container image.

Docker Hub
is the public repository of docker images

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docker hub includes images produce by docker certified images belonging to the docker Trusted registry and thousands of other images.

Docker Registry
is a scalable, open source storage and distribution system or docker image.

- Benefits of Docker
- Docker has revolutionerize the way we develop, ship and run appli^n by enabling containerization
- The key benefits are docker are:

1) Portability
- consistant environment: docker ensures that application run the same way across different environments.
- cross platform compatibility: docker containe can run on any system that supports docker.

2) Efficiency:
- lightweight: docker containers shares the host system os kernal making them more light weight.
- fast startup time: container can be started and stop quickly.

3) Scalability & performance:
- efficient resource utilization: unlike vm's containers do not require a full os to be replecated.

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• easy scaling: scalling can be done easily by running multiple instance of container.

4) Isolation
• Application isolation: each application runs in own isolated environment and do not interfare with each other.
• process Isolation: processes running inside a container are isolated from those running own the host or other containers.

5) Continues integration & continues deployment
• Integration with CI/CD tools
• Rapid and reliable deployment

• Microservice architecture
service decupling: docker facilities development and deployment of microservices by allowing each service to run in its own container.

6) Security
containers provide a level of security through isolation ensuring that application and their dependancy are reparated from the host system and each other.

• KUberNetice
it is the backbone of model contain orchastration enabling scalable, resilant and automated application deployment.
it is an open source orchastration tool developed by google.

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It helps, manage containerize applications in different deployment environment physical virtual and cloud.

- Features
  - High availability in down time
  - scalability or high performance
  - disaster, recovery backup and restore

- Components
  1) Pods : smallest unit of ubernatice
  - Abstraction over container
  - each pods has its own ip address
  - There is one appl per pod

2) Service
  - each appl is in service which has a permanent ip address
  - lif cycle of pod and service are not connected
  - There are two type of services
  1) external for an application so we can access it through the browser
  2) internal for database that shouldn't be exposed

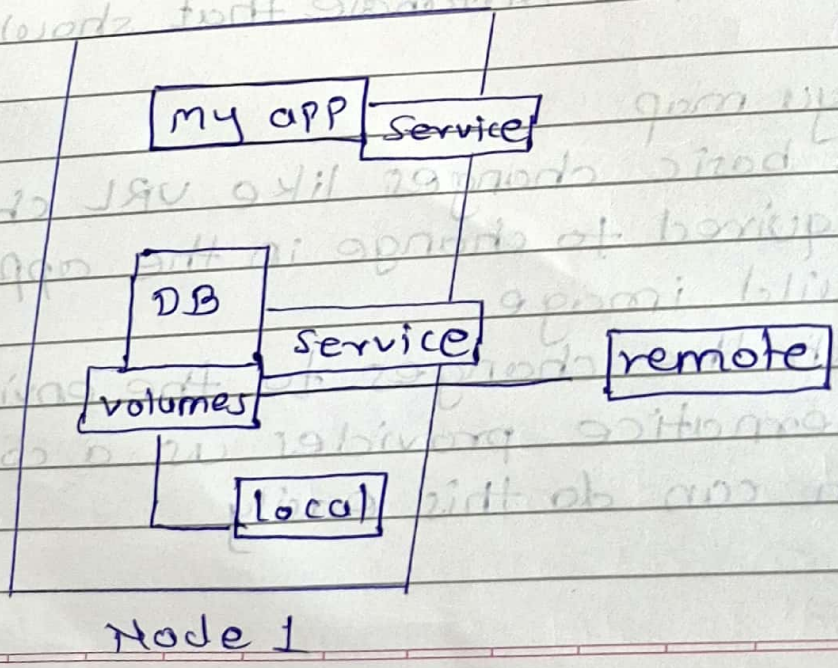
3) Configu map
  - For basic changes like URL changes could be required to change in the appl properties
  - Rebuild image
  - Push the changes to the environment
  - ubernatice provides us a configu map which can do this easily

4) secret : it is use to source secret data it is base 64 encoded

5) volumes: if your database crashes data is lost. ubernatice doesn't manage data persistancy. we can use volumes as local machine or external hard device drives. we as users are responcible for persistancy and backup

6) deployements: if one database or appi crashes their could be down time. in this case we need to specify the number of backup or container we want.
- The service we will point the same ip address.
- Helps in load balancing.

7) Stateful set
we can deploye data base directly because multiple containers could be writing to the same database for this we manage a stateful set.



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- Kubernetice Architecture

Worker processes:
we need three process for the worker machine in ~~Kubernetice~~ to work

- kubelet

The kubelet is the primary node agent that ensures containers are running in a pod as specified by the kubernetice control plane.

1) Functions:
- it receives instructures from the API server an ensures that the container describe in those specifications are running and healthy on its node.

2) interaction:
in the kubelet interact with the container runtime the container runtime interface manage container lifecycle image pooling and health checks.

3) Role: it reports the states of the node and its running pods back to the control plane allowing the cluster manager to make inform decisions about resource allocation and sheduling

- kubeproxy

is a network proxy that runs on each node to implement the kubernetice service concept.

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1) Function: it maintance network rules to allow network communication to and from Pods within the cluster.

2) Role: it manages port forwarding and simple load balancing for services, ensuring that traffic to a stable service ip is routed to the correct dynamically changing routed pod ip

- Container runtime

The container runtime is the underline software that is responsible for executing containers and managing containe image on node.

function:
it pulls container image from a registry unpack them and runs the appln within an isolated environment managing resources like CPU and memory. using mechanisms like C groups and namespaces

Role:
it provides a necessary environment for the application workload to run efficiently and securly

- MasterNode & Processes:

The kubernetice masternode runs several four processes responsible for managing and maintaining the desired state of the entire cluster

This components at as the brain of the cluster while worker nodes runs the actual applications. The main masterprocesses are.

i) API server.

This is the frontend of the kubernetice control plane and the single entry point for all communication it exposes the kubernetice API, process reset request, barrigets them and stores the configuration data in etcd

ii) etcd

A consistant highely available key value store that acts as the clusters single source of truth. it stores all data configuration and the current state of the cluster such as node detail services and node status.

iii) shedular

The process forcher for newly created node that have no node assign and selects an appropriate node for them to run over on it makes decision based on various factors like resource requirements, hardware constain, policy and availability.

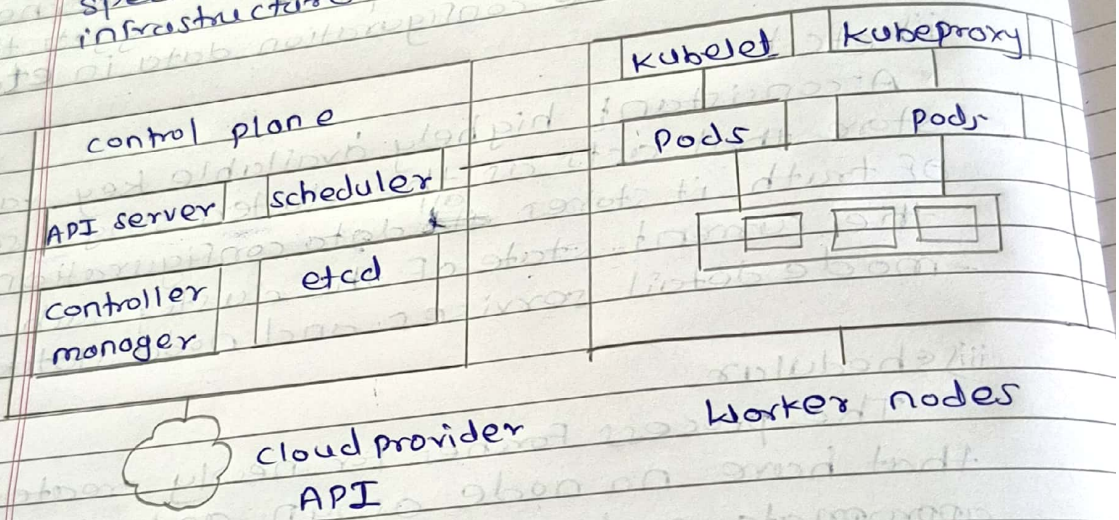
iv) controlel manager

kube controler manager is demon that runs various control loops that continuesly monitor the actual state of the cluster the api server and work to more it towards the desired state.

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Page No.
Date

cloud controller manager:
This component is used in cloud environments to manage interaction with the underline cloud providers API for task such as managing load balancers and storage volumes. it only runs controls specific to the cloud provider allowing cloud infrastructure to integrit with kubernetes.



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